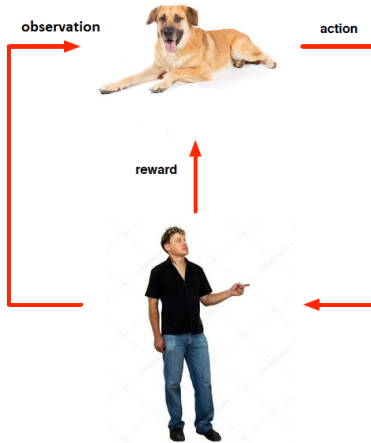


Lecture 1: Introduction to Reinforcement Learning. Dynamic Programming

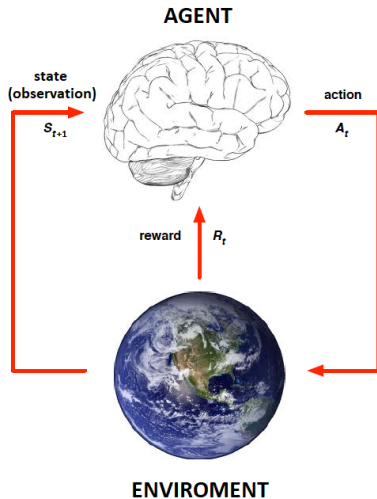
Anton Plaksin

What is Reinforcement Learning?

What is Reinforcement Learning?

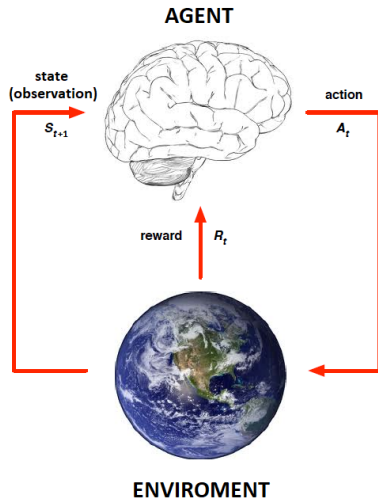


What is Reinforcement Learning?



The agent's goal is ???

What is Reinforcement Learning?



The agent's goal is to maximize $G = \sum_{t=0}^{\infty} \gamma^t R_t$, $\gamma \in [0, 1]$.

Markov Property

$$\mathbb{P}[S_{t+1}|S_t, A_t] = \mathbb{P}[S_{t+1}|S_0, A_0, S_1, A_1 \dots, S_t, A_t]$$

$$\mathbb{P}[R_t|S_t, A_t] = \mathbb{P}[R_t|S_0, A_0, S_1, A_1 \dots, S_t, A_t] = 1$$

Markov Decision Process

Markov Property

$$\mathbb{P}[S_{t+1}|S_t, A_t] = \mathbb{P}[S_{t+1}|S_0, A_0, S_1, A_1 \dots, S_t, A_t]$$

$$\mathbb{P}[R_t|S_t, A_t] = \mathbb{P}[R_t|S_0, A_0, S_1, A_1 \dots, S_t, A_t] = 1$$

Markov Decision Process $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{P}_0, \mathcal{R}, \gamma \rangle$

- ▶ $\mathcal{S}$ is a state space
- ▶ $\mathcal{A}$ is an action space
- ▶ $\mathcal{P}$ is a transition probability function

$$\mathcal{P}(s'|s, a) = \mathbb{P}[S_{t+1} = s'|S_t = s, A_t = a]$$

- ▶ $\mathcal{P}_0$ — an initial state probability function
- ▶ $\mathcal{R}$ — a reward function

$$\mathcal{R}(s, a) = R_t \quad \Leftrightarrow \quad \mathbb{P}[R_t|S_t = s, A_t = a] = 1$$

- ▶ $\gamma \in [0, 1]$ — discount coefficient

Markov Decision Process $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{P}_0, \mathcal{R}, \gamma \rangle$

The agent's goal is to maximize

$$G = \sum_{t=0}^{\infty} \gamma^t R_t$$

MDP with final states

Markov Decision Process $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{P}_0, \mathcal{R}, \gamma \rangle$

The agent's goal is to maximize

$$G = \sum_{t=0}^{\infty} \gamma^t R_t$$

Markov Decision Process $\langle \mathcal{S}, \mathcal{S}_F, \mathcal{A}, \mathcal{P}, \mathcal{P}_0, \mathcal{R}, \gamma \rangle$

► $\mathcal{S}_F$ — a set of final states

The agent's goal is to maximize

$$G = \sum_{t=0}^T \gamma^t R_t, \quad \text{if } S_T \in \mathcal{S}_F$$

or

$$G = \sum_{t=0}^{\infty} \gamma^t R_t \quad \text{if } S_t \notin \mathcal{S}_F$$

MDP with final states

Markov Decision Process $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{P}_0, \mathcal{R}, \gamma \rangle$

The agent's goal is to maximize

$$G = \sum_{t=0}^{\infty} \gamma^t R_t$$

Markov Decision Process $\langle \mathcal{S}, \mathcal{S}_F, \mathcal{A}, \mathcal{P}, \mathcal{P}_0, \mathcal{R}, \gamma \rangle$

► $\mathcal{S}_F$ — a set of final states

The agent's goal is to maximize

$$G = \sum_{t=0}^T \gamma^t R_t, \quad \text{if } S_T \in \mathcal{S}_F$$

or

$$G = \sum_{t=0}^{\infty} \gamma^t R_t \quad \text{if } S_t \notin \mathcal{S}_F$$

Are these statements equivalent?

Example: Multi-Armed Bandit

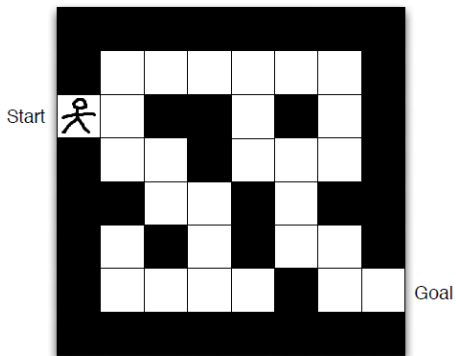


Example: Multi-Armed Bandit

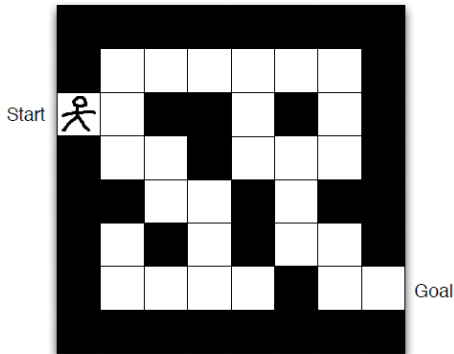


- ▶ States: one state
- ▶ Actions: to choose an arm
- ▶ Rewards: points
- ▶ No final state

Example: Maze



Example: Maze



- States: white cells
- Actions: $\uparrow$, $\rightarrow$, $\downarrow$, $\leftarrow$
- Rewards: -1 for each step, 0 if Goal
- Final state: Goal cell

Example: Multi-Armed Bandit



Example: Multi-Armed Bandit



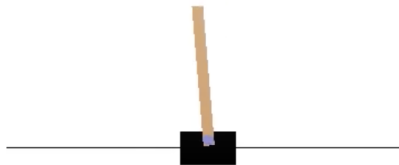
- States: one state
- Actions: to choose an arm
- Rewards: points

Example: Breakout Atari Game

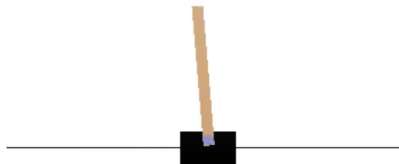


- ▶ States: pixels
- ▶ Actions: $\rightarrow$, $\leftarrow$, $-$
- ▶ Rewards: points
- ▶ Final states: when the ball falls down

Example: Cartpole



Example: Cartpole



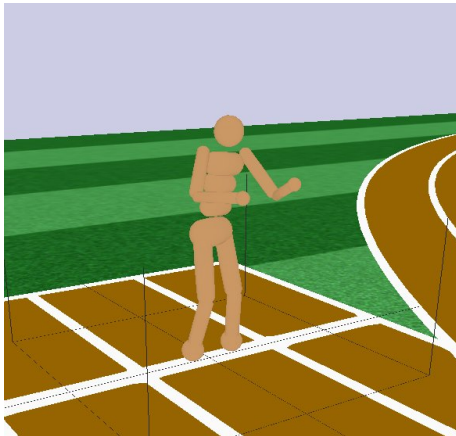
- ▶ States: $\mathbb{R}^4$
- ▶ Actions: $\rightarrow$, $\leftarrow$, 0
- ▶ Rewards: +1 for each step
- ▶ Final states: when the pole falls down

Example: Pendulum



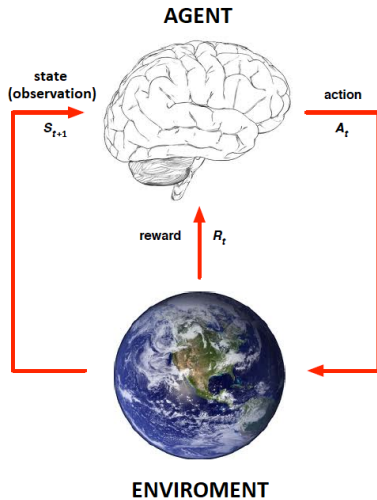
- ▶ State space: $\mathbb{R}^2$
or screen pixels
- ▶ Action space: $[-2, 2]$
- ▶ Rewards:
 $-\psi^2 - 0.1\dot{\psi}^2 - 0.001a^2$

Example: Humanoid



- ▶ States: $\mathbb{R}^{26}$
- ▶ Actions: $\mathbb{R}^6$
- ▶ Rewards: +1 for each step
- ▶ Final states: when the Humanoid falls down

What we want to find?



$$\pi: \mathcal{S} \mapsto \mathcal{A}$$

$$\pi: \mathcal{S} \mapsto \mathcal{A}$$

► Set π

$$\pi: \mathcal{S} \mapsto \mathcal{A}$$

- ▶ Set π
- ▶ Agent starts from the initial state $S_0 \sim \mathcal{P}_0$

$$\pi: \mathcal{S} \mapsto \mathcal{A}$$

- ▶ Set π
- ▶ Agent starts from the initial state $S_0 \sim \mathcal{P}_0$
- ▶ acts $A_0 = \pi(S_0)$

$$\pi: \mathcal{S} \mapsto \mathcal{A}$$

- ▶ Set π
- ▶ Agent starts from the initial state $S_0 \sim \mathcal{P}_0$
- ▶ acts $A_0 = \pi(S_0)$
- ▶ gets the reward $R_0 = \mathcal{R}(S_0, A_0)$ and the next state $S_1 \sim \mathcal{P}(\cdot | S_0, A_0)$

$$\pi: \mathcal{S} \mapsto \mathcal{A}$$

- ▶ Set π
- ▶ Agent starts from the initial state $S_0 \sim \mathcal{P}_0$
- ▶ acts $A_0 = \pi(S_0)$
- ▶ gets the reward $R_0 = \mathcal{R}(S_0, A_0)$ and the next state $S_1 \sim \mathcal{P}(\cdot | S_0, A_0)$
- ▶ acts $A_1 = \pi(S_1)$

$$\pi: \mathcal{S} \mapsto \mathcal{A}$$

- ▶ Set π
- ▶ Agent starts from the initial state $S_0 \sim \mathcal{P}_0$
- ▶ acts $A_0 = \pi(S_0)$
- ▶ gets the reward $R_0 = \mathcal{R}(S_0, A_0)$ and the next state $S_1 \sim \mathcal{P}(\cdot|S_0, A_0)$
- ▶ acts $A_1 = \pi(S_1)$
- ▶ gets the reward $R_1 = \mathcal{R}(S_1, A_1)$ and the next state $S_2 \sim \mathcal{P}(\cdot|S_1, A_1)$

$$\pi: \mathcal{S} \mapsto \mathcal{A}$$

- ▶ Set π
- ▶ Agent starts from the initial state $S_0 \sim \mathcal{P}_0$
- ▶ acts $A_0 = \pi(S_0)$
- ▶ gets the reward $R_0 = \mathcal{R}(S_0, A_0)$ and the next state $S_1 \sim \mathcal{P}(\cdot | S_0, A_0)$
- ▶ acts $A_1 = \pi(S_1)$
- ▶ gets the reward $R_1 = \mathcal{R}(S_1, A_1)$ and the next state $S_2 \sim \mathcal{P}(\cdot | S_1, A_1)$
- ▶ ...
- ▶ $\tau = \{S_0, A_0, S_1, A_1, S_2, A_2, \dots\}$, $G(\tau) = \sum_{t=0}^{\infty} \gamma^t \mathcal{R}(S_t, A_t)$

$$\pi: \mathcal{S} \mapsto \mathcal{A}$$

- ▶ Set π
- ▶ Agent starts from the initial state $S_0 \sim \mathcal{P}_0$
- ▶ acts $A_0 = \pi(S_0)$
- ▶ gets the reward $R_0 = \mathcal{R}(S_0, A_0)$ and the next state $S_1 \sim \mathcal{P}(\cdot|S_0, A_0)$
- ▶ acts $A_1 = \pi(S_1)$
- ▶ gets the reward $R_1 = \mathcal{R}(S_1, A_1)$ and the next state $S_2 \sim \mathcal{P}(\cdot|S_1, A_1)$
- ▶ ...
- ▶ $\tau = \{S_0, A_0, S_1, A_1, S_2, A_2, \dots\}$, $G(\tau) = \sum_{t=0}^{\infty} \gamma^t \mathcal{R}(S_t, A_t)$

The Reinforcement Learning problem

$$\mathbb{E}_{\tau \sim \pi} [G(\tau)] \longrightarrow \max_{\pi}$$

$$\pi(a|s) \in [0, 1], \quad a \in \mathcal{A}, \quad s \in \mathcal{S}$$

- ▶ Set π
- ▶ Agent starts from the initial state $S_0 \sim \mathcal{P}_0$
- ▶ acts $A_0 \sim \pi(\cdot|S_0)$
- ▶ gets the reward $R_0 = \mathcal{R}(S_0, A_0)$ and the next state $S_1 \sim \mathcal{P}(\cdot|S_0, A_0)$
- ▶ acts $A_1 \sim \pi(\cdot|S_1)$
- ▶ gets the reward $R_1 = \mathcal{R}(S_1, A_1)$ and the next state $S_2 \sim \mathcal{P}(\cdot|S_1, A_1)$
- ▶ ...
- ▶ $\tau = \{S_0, A_0, S_1, A_1, S_2, A_2, \dots\}, \quad G(\tau) = \sum_{t=0}^{\infty} \gamma^t \mathcal{R}(S_t, A_t)$

The Reinforcement Learning problem

$$\mathbb{E}_{\tau \sim \pi} [G(\tau)] \longrightarrow \max_{\pi}$$

"Optimal" Policy

Theorem

Let $\mathcal{S}$ and $\mathcal{A}$ be finite. Let $\mathcal{R}$ be bounded. Let $\gamma \in (0, 1)$.
Then there exists a policy π_* such that

$$\mathbb{E}_{\tau \sim \pi_*} [G(\tau)] \geq \mathbb{E}_{\tau \sim \pi} [G(\tau)], \quad \forall \pi$$

Value Function

- ▶ Set π and s
- ▶ Agent starts from the initial state $S_0 = s$
- ▶ acts $A_0 \sim \pi(\cdot|S_0)$
- ▶ gets the reward $R_0 = \mathcal{R}(S_0, A_0)$ and the next state $S_1 \sim \mathcal{P}(\cdot|S_0, A_0)$
- ▶ acts $A_1 \sim \pi(\cdot|S_1)$
- ▶ gets the reward $R_1 = \mathcal{R}(S_1, A_1)$ and the next state $S_2 \sim \mathcal{P}(\cdot|S_1, A_1)$
- ▶ ...
- ▶ $\tau = \{S_0, A_0, S_1, A_1, S_2, A_2, \dots\}, \quad G(\tau) = \sum_{t=0}^{\infty} \gamma^t \mathcal{R}(S_t, A_t)$

Value Function

- ▶ Set π and s
- ▶ Agent starts from the initial state $S_0 = s$
- ▶ acts $A_0 \sim \pi(\cdot|S_0)$
- ▶ gets the reward $R_0 = \mathcal{R}(S_0, A_0)$ and the next state $S_1 \sim \mathcal{P}(\cdot|S_0, A_0)$
- ▶ acts $A_1 \sim \pi(\cdot|S_1)$
- ▶ gets the reward $R_1 = \mathcal{R}(S_1, A_1)$ and the next state $S_2 \sim \mathcal{P}(\cdot|S_1, A_1)$
- ▶ ...
- ▶ $\tau = \{S_0, A_0, S_1, A_1, S_2, A_2, \dots\}$, $G(\tau) = \sum_{t=0}^{\infty} \gamma^t \mathcal{R}(S_t, A_t)$

Value Function

$$v_{\pi}(s) = \mathbb{E}_{\tau \sim \pi} [G(\tau) \mid S_0 = s]$$

Partially Order for Policies

$$\pi' \geq \pi \quad \Leftrightarrow \quad v_{\pi'}(s) \geq v_{\pi}(s), \quad \forall s \in \mathcal{S}$$

Partially Order for Policies

$$\pi' \geq \pi \quad \Leftrightarrow \quad v_{\pi'}(s) \geq v_{\pi}(s), \quad \forall s \in \mathcal{S}$$

Theorem

Let $\mathcal{S}$ and $\mathcal{A}$ be finite. Let $\mathcal{R}$ be bounded. Let $\gamma \in (0, 1)$.
There exists a policy π_* such that

$$\pi^* \geq \pi, \quad \forall \pi$$

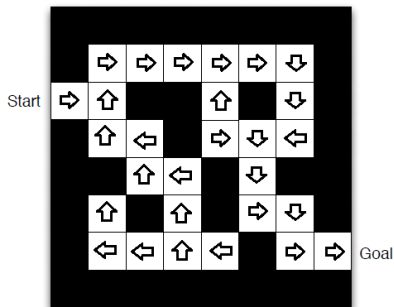
Deterministic Example: Maze

Remark

If Policy and Environment are deterministic (non-stochastic) then

$$v_{\pi}(s) = G(\tau_{\pi}), \quad \mathbb{P}(\tau_{\pi}|\pi) = 1.$$

$$R_t = -1, \quad \gamma = 1, \quad \pi :$$



Deterministic Example: Maze

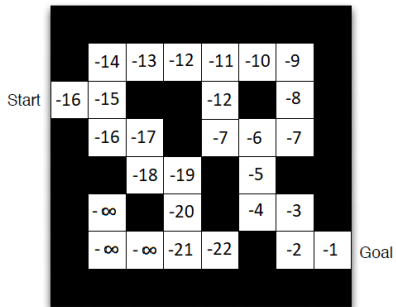
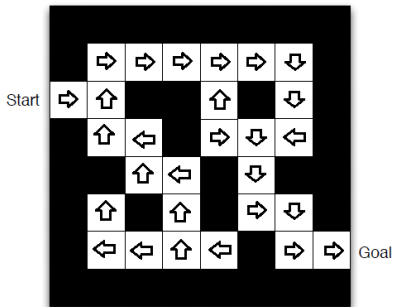
Remark

If Policy and Environment are deterministic (non-stochastic) then

$$v_\pi(s) = G(\tau_\pi), \quad \mathbb{P}(\tau_\pi|\pi) = 1.$$

$$R_t = -1, \quad \gamma = 1, \quad \pi :$$

$V_{\pi} :$



Bellman Expectation Equation (Deterministic Case)

$$S_0 = s \Rightarrow \tau = (S_0, A_0, S_1, A_1, \dots) \Rightarrow v_\pi(s) = G(\tau) = \sum_{t=0}^{\infty} \gamma^t \mathcal{R}(S_t, A_t)$$

Bellman Expectation Equation (Deterministic Case)

$$S_0 = s \Rightarrow \tau = (S_0, A_0, S_1, A_1, \dots) \Rightarrow v_\pi(s) = G(\tau) = \sum_{t=0}^{\infty} \gamma^t \mathcal{R}(S_t, A_t)$$

$$\tilde{\tau} = (S_1, A_1, S_2, A_2, S_3, A_3 \dots), \Rightarrow v_\pi(S_1) = G(\tilde{\tau}) = \sum_{t=0}^{\infty} \gamma^t \mathcal{R}(S_{t+1}, A_{t+1})$$

Bellman Expectation Equation (Deterministic Case)

$$S_0 = s \Rightarrow \tau = (S_0, A_0, S_1, A_1, \dots) \Rightarrow v_\pi(s) = G(\tau) = \sum_{t=0}^{\infty} \gamma^t \mathcal{R}(S_t, A_t)$$

$$\tilde{\tau} = (S_1, A_1, S_2, A_2, S_3, A_3, \dots), \Rightarrow v_\pi(S_1) = G(\tilde{\tau}) = \sum_{t=0}^{\infty} \gamma^t \mathcal{R}(S_{t+1}, A_{t+1})$$

$$v_\pi(s) = \mathcal{R}(S_0, A_0) + \gamma \sum_{t=1}^{\infty} \gamma^{t-1} \mathcal{R}(S_t, A_t) = \mathcal{R}(S_0, A_0) + \gamma v_\pi(S_1)$$

Bellman Expectation Equation (Deterministic Case)

$$S_0 = s \Rightarrow \tau = (S_0, A_0, S_1, A_1, \dots) \Rightarrow v_\pi(s) = G(\tau) = \sum_{t=0}^{\infty} \gamma^t \mathcal{R}(S_t, A_t)$$

$$\tilde{\tau} = (S_1, A_1, S_2, A_2, S_3, A_3, \dots), \Rightarrow v_\pi(S_1) = G(\tilde{\tau}) = \sum_{t=0}^{\infty} \gamma^t \mathcal{R}(S_{t+1}, A_{t+1})$$

$$v_\pi(s) = \mathcal{R}(S_0, A_0) + \gamma \sum_{t=1}^{\infty} \gamma^{t-1} \mathcal{R}(S_t, A_t) = \mathcal{R}(S_0, A_0) + \gamma v_\pi(S_1)$$

Bellman Expectation Equation for v_π

$$v_\pi(s) = \mathcal{R}(s, a) + \gamma v_\pi(s'), \quad \pi(a|s) = 1, \quad \mathcal{P}(s'|s, a) = 1$$

Bellman Expectation Equation (General Case)

Bellman Expectation Equation for v_π

$$v_\pi(s) = \sum_a \pi(a|s) \left(\mathcal{R}(s, a) + \gamma \sum_{s'} \mathcal{P}(s'|s, a) v_\pi(s') \right)$$

Action-Value (Q-) Function

- ▶ Set π and s, a
- ▶ Agent starts from the initial state $S_0 = s$
- ▶ acts $A_0 = a$
- ▶ gets the reward $R_0 = \mathcal{R}(S_0, A_0)$ and the next state $S_1 \sim \mathcal{P}(\cdot | S_0, A_0)$
- ▶ acts $A_1 \sim \pi(\cdot | S_1)$
- ▶ gets the reward $R_1 = \mathcal{R}(S_1, A_1)$ and the next state $S_2 \sim \mathcal{P}(\cdot | S_1, A_1)$
- ▶ ...
- ▶ $\tau = \{S_0, A_0, S_1, A_1, S_2, A_2, \dots\}, \quad G(\tau) = \sum_{t=0}^{\infty} \gamma^t \mathcal{R}(S_t, A_t)$

Action-Value (Q-) Function

- ▶ Set π and s, a
- ▶ Agent starts from the initial state $S_0 = s$
- ▶ acts $A_0 = a$
- ▶ gets the reward $R_0 = \mathcal{R}(S_0, A_0)$ and the next state $S_1 \sim \mathcal{P}(\cdot | S_0, A_0)$
- ▶ acts $A_1 \sim \pi(\cdot | S_1)$
- ▶ gets the reward $R_1 = \mathcal{R}(S_1, A_1)$ and the next state $S_2 \sim \mathcal{P}(\cdot | S_1, A_1)$
- ▶ ...
- ▶ $\tau = \{S_0, A_0, S_1, A_1, S_2, A_2, \dots\}, \quad G(\tau) = \sum_{t=0}^{\infty} \gamma^t \mathcal{R}(S_t, A_t)$

Q-Function

$$q_{\pi}(s, a) = \mathbb{E}_{\tau \sim \pi} [G(\tau) \mid S_0 = s, A_0 = a]$$

Action-Value Function

$$q_{\pi}(s, a) = \mathbb{E}_{\pi}[G \mid S_0 = s, A_0 = a]$$

Action-Value Function

$$q_{\pi}(s, a) = \mathbb{E}_{\pi}[G \mid S_0 = s, A_0 = a]$$

q_{π} and v_{π}

$$v_{\pi}(s) = \sum_a \pi(a|s) q_{\pi}(s, a), \quad q_{\pi}(s, a) = \mathcal{R}(s, a) + \gamma \sum_{s'} \mathcal{P}(s'|s, a) v_{\pi}(s')$$

Action-Value Function

$$q_{\pi}(s, a) = \mathbb{E}_{\pi}[G \mid S_0 = s, A_0 = a]$$

q_{π} and v_{π}

$$v_{\pi}(s) = \sum_a \pi(a|s)q_{\pi}(s, a), \quad q_{\pi}(s, a) = \mathcal{R}(s, a) + \gamma \sum_{s'} \mathcal{P}(s'|s, a)v_{\pi}(s')$$

Bellman Expectation Equation for q_{π}

$$q_{\pi}(s, a) = \mathcal{R}(s, a) + \gamma \sum_{s'} \mathcal{P}(s'|s, a) \sum_{a'} \pi(a'|s')q_{\pi}(s', a')$$

QUESTIONS?